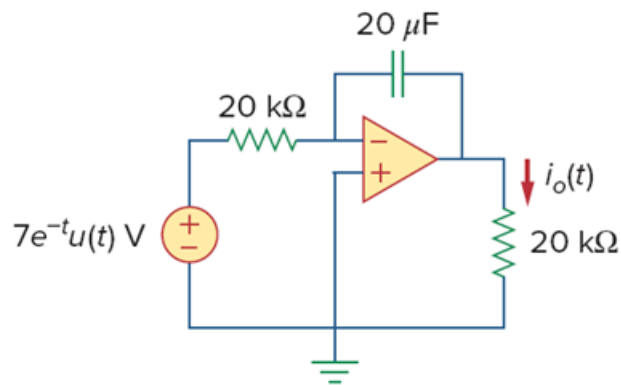


Problem

Find $i_o(t)$ in the op amp circuit of Fig. 18.50.

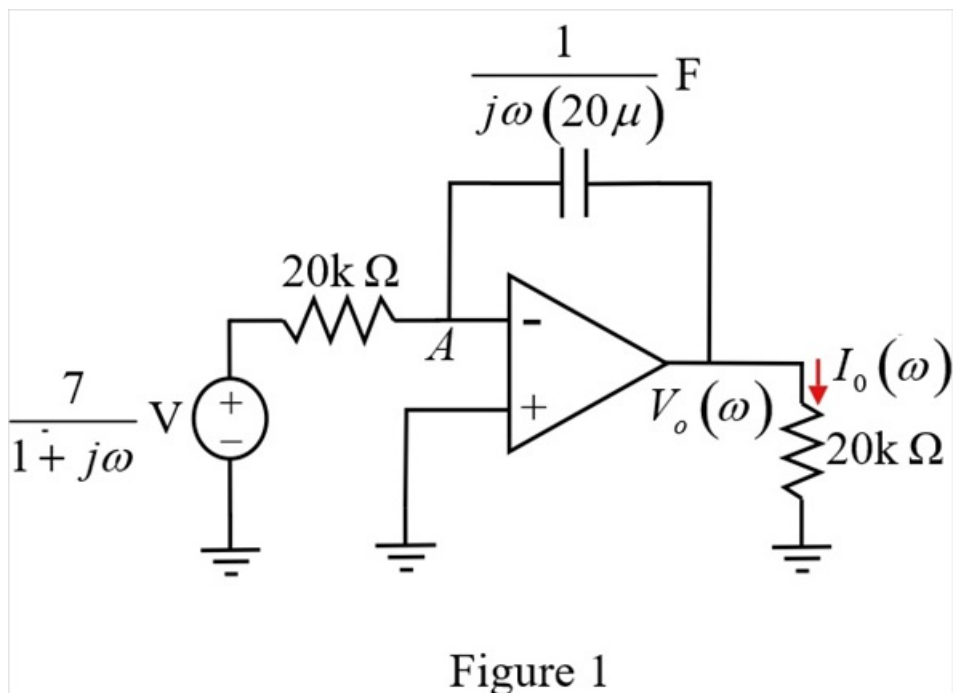
Figure 18.50



Step-by-step solution

Step 1 of 5

(Refer to Figure 18.50 in the textbook) represent the circuit in frequency domain.



Step 2 of 5

The voltage at node A is,

$$V_A = 0 \text{ V (Due to the virtual ground).}$$

Apply the nodal analysis at node A .

$$\begin{aligned} 0 - \left(\frac{7}{1 + j\omega} \right) \frac{1}{20 \text{ k}} + \frac{0 - V_o(\omega)}{\left(\frac{1}{j\omega(20\mu)} \right)} &= 0 \\ - \left(\frac{7}{1 + j\omega} \right) &= V_o(\omega) j\omega(20\mu)(20 \text{ k}) \\ V_o(\omega) &= \frac{-17.5}{j\omega(1 + j\omega)} \end{aligned}$$

Step 3 of 5

The current across the resistor is,

$$I_o(\omega) = \frac{V_o(\omega)}{20 \text{ k}}$$

Substitute $\frac{-17.5}{j\omega(1 + j\omega)}$ for $V_o(\omega)$.

$$\begin{aligned} I_o(\omega) &= \frac{\left(\frac{-17.5}{j\omega(1 + j\omega)} \right)}{20 \text{ k}} \\ &= -0.875 \left(\frac{1}{j\omega(1 + j\omega)} \right) \text{ mA} \dots\dots (1) \end{aligned}$$

Step 4 of 5

Calculate the partial fraction of equation (1).

$$\frac{1}{j\omega(1 + j\omega)} = \frac{A}{j\omega} + \frac{B}{1 + j\omega}$$

Substitute 1 for A and -1 for B .

$$\frac{1}{j\omega(1 + j\omega)} = \frac{1}{j\omega} - \frac{1}{1 + j\omega}$$

Rewrite the equation (1).

$$I_o(\omega) = \frac{0.875}{1 + j\omega} - \frac{0.875}{j\omega}$$

Step 5 of 5

Calculate the inverse Fourier transform.

$$\begin{aligned} F^{-1}[I_o(\omega)] &= F^{-1}\left(\frac{0.875}{1 + j\omega}\right) - F^{-1}\left(\frac{0.875}{j\omega}\right) \\ i_o(t) &= (0.875e^{-t} - 0.875)u(t) \text{ mA} \end{aligned}$$

Hence, the current across the resistor is $\boxed{(0.875e^{-t} - 0.875)u(t) \text{ mA}}$.